# SRM Institute of Science and Technology

**(Deemed to be University U/S 3 of UGC Act, 1956)**

## Faculty of Science and Humanities

**Tiruchirappalli, Tamil Nadu-621 105**

---

## Department of Computer Applications

PRACTICAL RECORD

NAME :

REGISTER NO :

COURSE : MCA (General)

SEMESTER/ YEAR : I / I

SUBJECT CODE : PCA25C01J

SUBJECT NAME : Object Oriented Programming Using Java

**November 2025**

# SRM Institute of Science and Technology

**(Deemed to be University U/S 3 of UGC Act, 1956)**

## Faculty of Science and Humanities

**Tiruchirappalli, Tamil Nadu-621 105**

---

# Department of Computer Applications

## BONAFIDE CERTIFICATE

This is to certify that the bonafide work is done by Mr/Ms. _____ Register No. _____ in **Object Oriented Programming Using Java (Subject Code: PCA25C01J)** at Computer Lab, SRMIST, Tiruchirappalli, in November 2025.

**STAFF IN-CHARGE** **HEAD OF THE DEPARTMENT**

Submitted for the University Practical Examination held at SRMIST, Tiruchirappalli, Department of Computer Applications on _____.

**INTERNAL EXAMINER** **EXTERNAL EXAMINER**

**INDEX**


| S. No. | Date | Name of the Program | Page No. | Signature |
|---|---|---|---|---|
| 1 | 16.07.2025 | JAVA INPUT AND OUTPUT | | |
| 2 | 23.07.2025 | TYPE CONVERSION | | |
| 3 | 30.07.2025 | JAVA OPERATORS | | |
| 4 | 13.08.2025 | LOOPING STATEMENTS | | |
| 5 | 20.08.2025 | WHILE STATEMENT | | |
| 6 | 20.08.2025 | DO-WHILE STATEMENT | | |
| 7 | 27.08.2025 | ARRAYS | | |
| 8 | 03.09.2025 | SWITCH-CASE STATEMENT | | |
| 9 | 10.09.2025 | COMMAND LINE ARGUMENTS AND STRING FUNCTIONS | | |
| 10 | 17.09.2025 | INHERITANCE | | |
| 11 | 24.09.2025 | EXCEPTIONS | | |
| 12 | 01.10.2025 | MULTI-THREADING | | |
| 13 | 01.10.2025 | PACKAGE | | |
| 14 | 08.10.2025 | INTERFACE | | |
| 15 | 08.10.2025 | LAYOUT MANAGER (FLOW LAYOUT) | | |

**Ex. No. 1** **JAVA INPUT AND OUTPUT**

**Date:** 16.07.2025

**AIM:**

To create a java program to display the name of the institution, year of establishment, department and your class.

**ALGORITHM:**

Step 1: Start the Program

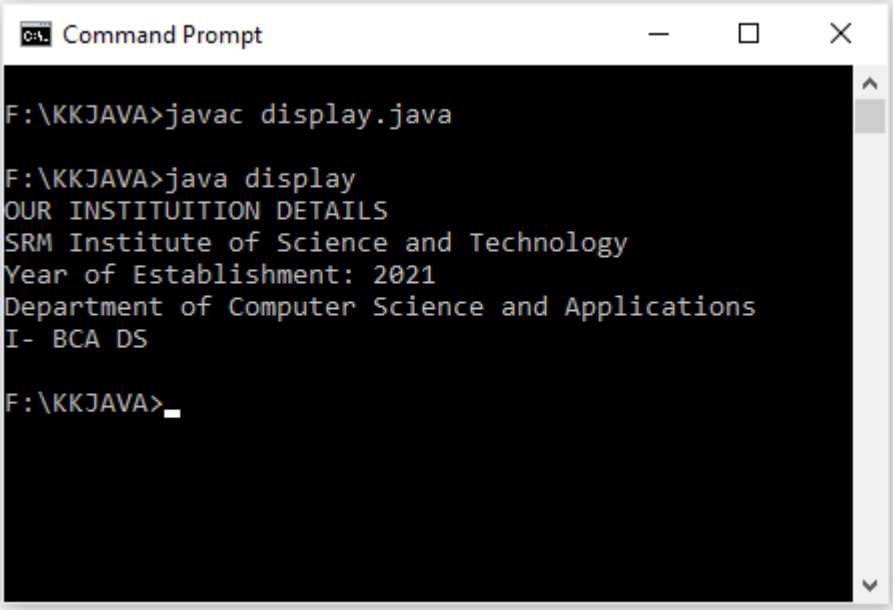
Step2: Define a class and display name of the institution, year of establishment, department and your class using Output method

Step3: Stop the program.

**PROGRAM:**

```
import java.io.*;
class display
{
public static void main(String args[])
{
int year=2021;
System.out.println("OUR INSTITUITION DETAILS");
System.out.println("SRM Institute of Science and Technology");
System.out.println("Year of Establishment: " + year);
System.out.println("Department of Computer Applications");
System.out.println("I- MCA-GENERAL");
}
}
```

**OUTPUT:**




**RESULT:**

The program runs successfully, displaying the institution's details as specified.

**Ex. No. 2** **TYPE CONVERSION**

**Date:** 23.07.2025

**AIM:**

To create a java program that performs type conversion.

**ALGORITHM:**

Step 1: Start the program.

Step 2: Define a class and its methods.

Step 3: Declare and initialize variables ; i=100. Step 4:

Perform type conversion like long and float

Long l=i, float f=I;

Step 5: Display the result.

Step 6: Stop the program.

**PROGRAM:**

```
// Automatic type conversion program

import java.io.*;
class conversion
 {
    public static void main(String[] args)
   {
      int i = 100;

     // Integer to long type
      long l = i;

       // long to float type
      float f = l;

      // Print and display commands
      System.out.println("Int value " + i);
      System.out.println("Long value " + l);
      System.out.println("Float value " + f);
   }
}
```

**OUTPUT:**

Command Prompt

```
F:\KKJAVA>javac conversion.java

F:\KKJAVA>java conversion
Int value 100
Long value 100
Float value 100.0

F:\KKJAVA>
```

**RESULT:**

The values have been successfully converted and displayed.

**Ex. No. 3** **JAVA OPERATORS**

**Date:** 30.07.2025

**AIM:**

To create a Java program to implement Operators.

**ALGORITHM:**

Step 1: Start the Program

Step 2: Define a class and declare variables like a,b,add, sub, mul, div, moddiv;

Step 3 : assign value to the variables a=10, b=20;

Step 4: perform arithmetic operations add=a+b, sub= b-a, mul=a*b, div=a/2, moddiv=a%3

Step 5: Display the results

Step 6: Stop the program.

**PROGRAM:**

```
import java.io.*;
class operators
{
public static void main(String args[])
{
        int a=100,b=200,add, sub,
        mul,div; add=a+b;
        sub=b-a;
        mul=a*b;
        div=a/b;
        System.out.println("ARITHMETIC OPERATORS:");
        System.out.println("Addition:" + add);
        System.out.println("Subtraction:"+ sub);
        System.out.println("Multiplication:"+ mul);
        System.out.println("Division"+ div); System.out.println("Unary
        Operators: ++, --"); System.out.println(a++);
        System.out.println(++a);
        System.out.println(a--)
        ;
        System.out.println(--a)
        ;
        System.out.println("SHIFT OPERATORS");
        System.out.println(10<<2);
        System.out.println(10<<3);
        System.out.println(20<<2);
        System.out.println(15<<4);
}}
```

**OUTPUT:**

```
Command Prompt

F:\KKJAVA>javac operators.java

F:\KKJAVA>java operators
ARITHMETIC OPERATORS:
Addition:300
Subtraction:100
Multiplication:20000
Division0
Unary Operators: ++, --
100
102
102
100
SHIFT OPERATORS
40
80
80
240

F:\KKJAVA>
```

**RESULT:**

The program demonstrates various operators in Java, including arithmetic, unary, and shift operators and runs successfully.

**Ex. No. 4** **LOOPING STATEMENTS**

**Date:** 13.08.2025

**AIM:**

To create a Java program to implement looping statements

**ALGORITHM:**

Step 1: Start the Program

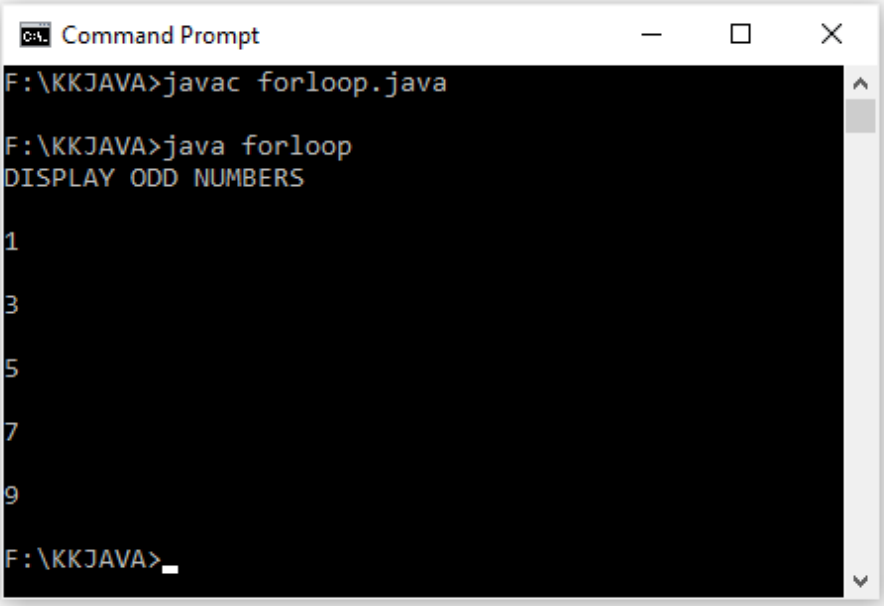
Step2: Define a class to display odd numbers using for-loop

Step3: Stop the program.

**PROGRAM:**

```
// FOR-LOOP : DISPLAY ODD NUMBER
import java.io.*;
public class forloop
 {
public static void main(String args[ ])
 {
System.out.println(“ DISPLAY ODD NUMBERS");
for(int i=1;i<=10;i=i+2)
   {
System.out.println("\n"+i);
    }
 }
  }
```

**OUTPUT:**




**RESULT:**

The program uses a for loop to display odd numbers from 1 to 10 and runs successfully.

**Ex. No. 5** **WHILE STATEMENT**

**Date:** 20.08.2025

**AIM:**

To write a java program to implement while-looping statements

**ALGORITHM:**

Step 1: Start the Program

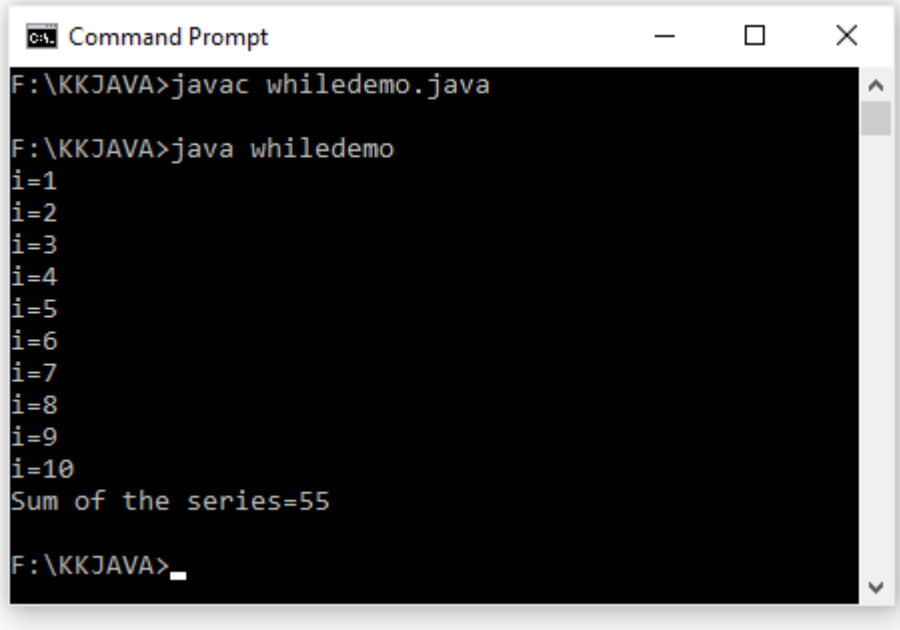
Step 2: Define a class to find the sum of the series 1+2+3+ …. +10 using a while-loop

Step3: Stop the program

**PROGRAM:**

```
import java.io.*; class
whiledemo
{
public static void main(String args[] )
{
int i=1, sum=0;
while(i<=10)
   {
System.out.println("i="+i);
sum=sum+i;
i++;
}
System.out.println("Sum of the series="+sum);
  }

}
```

**OUTPUT:**




**RESULT:**

The program uses a while loop to calculate the sum of the series 1+2+3+...+10 and runs successfully.

**Ex. No. 6** **DO-WHILE STATEMENT**

**Date:** 20.08.2025

**AIM:**

To write a java program to implement Do-while looping statements

**ALGORITHM:**

Step 1: Start the Program

Step2: Define a class to Print the even numbers from 2 to 20 using D-while-loop. Step3:

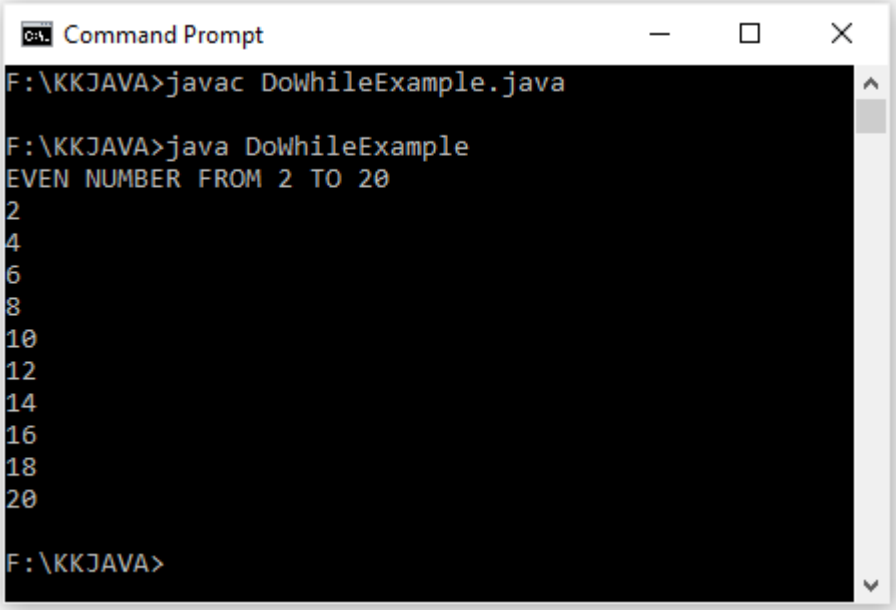
Stop the program

**PROGRAM:**

**// DoWhileExample.java to Print the even numbers from 2 to 20**

```
public class DoWhileExample
 {
public static void main(String args[] )
{
   int i=2;
   System.out.println("EVEN NUMBER FROM 2 TO 20");
   do
  {
    System.out.println(i); i=i+2;
   }
   while(i<=20);
  }
}
```

**OUTPUT:**




**RESULT:**

The program uses a do-while loop to print the even numbers from 2 to 20 and program runs successfully.

**Ex. No. 7** **ARRAYS**

**Date:** 27.08.2025

**AIM:**

To implement array concept using Java

**ALGORITHM:**

Step 1: Start the program

Step 2: Create a class

Step 3: Declare 2D array variables

Step 4: Declare methods with for loop to give input

Step 5: Perform Matrix Addition

Step 6: Display the output

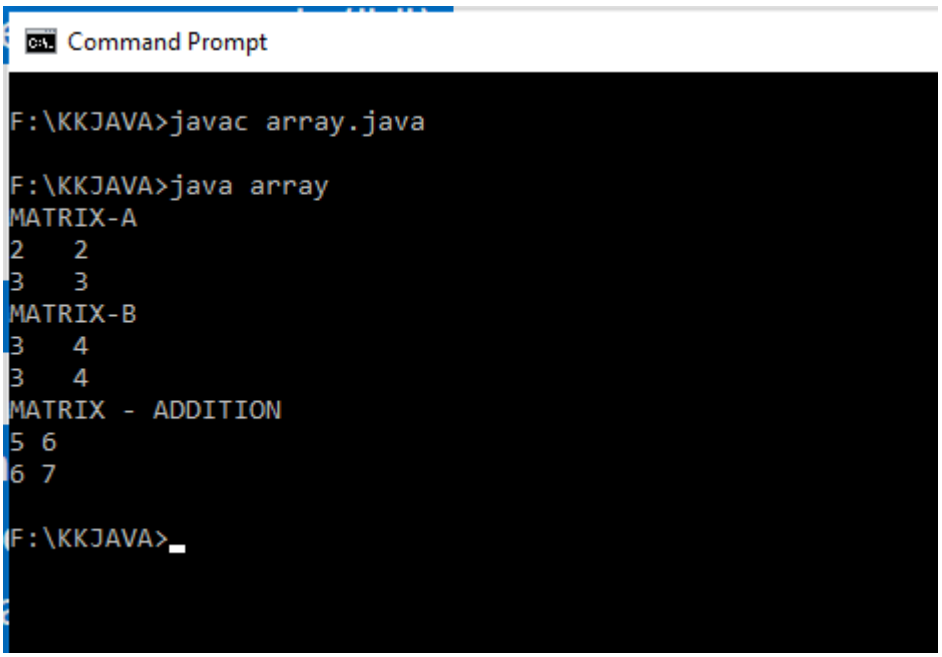
Step 7: Stop the program

**PROGRAM:**

```
//MATRIX ADDITION
import java.io.*;

class array
{

inta[][]= new int[2][2];
intb[][]=new int[2][2];
intc[][]=new int[2][2]; inti,j;

void get()
{
System.out.println("MATRIX-A")
; for(i=0;i<a.length;i++)
{
for(j=0;j<a.length;j++)
{
a[i][j]=i+2;
System.out.print(a[i][j]+" ");
}
System.out.println(" ");
}
System.out.println("MATRIX-B");
for(i=0;i<b.length;i++)
{
for(j=0;j<b.length;j++)
{
b[i][j]=j+3;
System.out.print(b[i][j]+" ");
}
System.out.println(" ");
}
}

void cal()
 {
for(i=0;i<a.length;i++
)
for(j=0;j<b.length;j++
)
{
```

```
c[i][j]=a[i][j]+b[i][j];
}
}

void disp()
{
System.out.println("MATRIX - ADDITION"); for(i=0;i<a.length;i++)
{
for(j=0;j<b.length;j++)
{
System.out.print(c[i][j]+" ");
}
System.out.println(" ");
}
}

public static void main(String args[])
{
 array ar=new
array(); ar.get();
ar.cal();
ar.disp();

}
}
```

**OUTPUT:**




**RESULT:**

The program performs matrix addition on two 2x2 matrices A and B, resulting matrix C (Matrix Addition) runs successfully.

# SWITCH-CASE STATEMENT

**Ex. No. 8**
**Date:** 03.09.2025

**AIM:**

To implement switch-case statements using Java.

**ALGORITHM:**

Step 1: Start the program

Step 2: Read days of a week

Step 3: Create switch with multiple statements

Step 4: Enter your choice

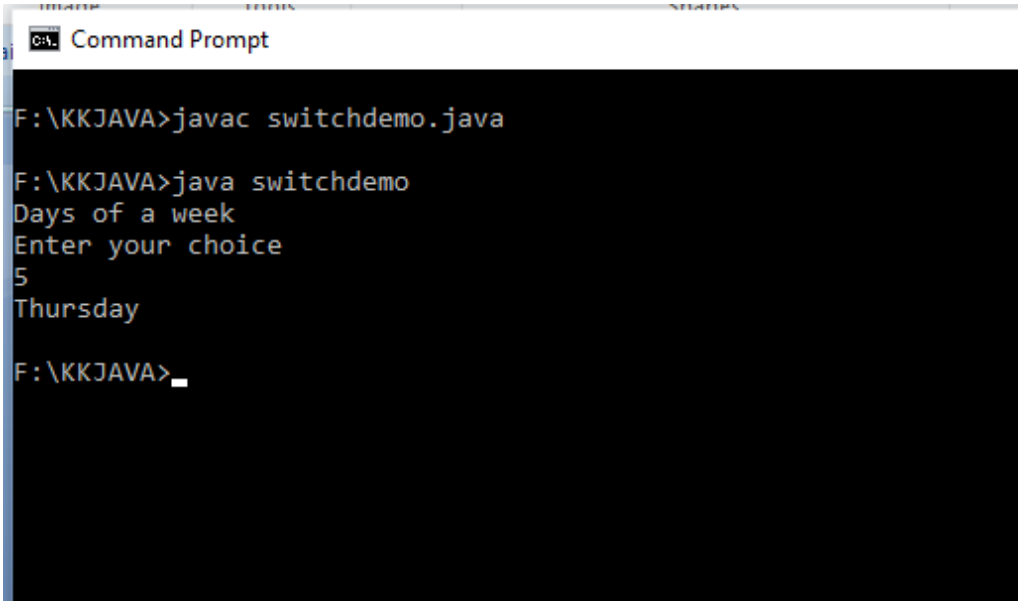
Step 5: Display the day as output

Step 6: stop the program.

**PROGRAM:**

```
import java.util.*; // Use java.util for Scanner

class SwitchDemo {
   public static void main(String[] args) {
      int day;
      Scanner console = new Scanner(System.in);

      System.out.println("Days of a Week");
      System.out.println("Enter your choice (1–7):");
      day = console.nextInt();

      switch (day) {
         case 1:
            System.out.println("Sunday");
            break;
         case 2:
            System.out.println("Monday");
            break;
         case 3:
            System.out.println("Tuesday");
            break;
         case 4:
            System.out.println("Wednesday");
            break;
         case 5:
            System.out.println("Thursday");
            break;
         case 6:
            System.out.println("Friday");
            break;
         case 7:
            System.out.println("Saturday");
            break;
         default:
            System.out.println("Invalid choice! Please enter a number between 1 and 7.");
      }

      console.close();
   }
```

**OUTPUT:**




**RESULT:**

The switch-case execution program has been successfully demonstrated in Java.

# COMMAND-LINE ARGUMENTS AND STRING FUNCTIONS

**Ex. No. 9**
**Date:** 10.09.2025

**AIM:**

To implement command line arguments and string functions using Java.

**ALGORITHM:**

Step 1: Start the program

Step 2: Declare variables with 3 data types int, float and string.

Step 3: Get 3 inputs from the command line during runtime as given below. java cmdline 12 15.5 AMMU

Step 4: Convert numeric input to respective data types.

Step 5: Perform Calculation and display results of numeric input.

Step 6: Implement String functions on the given input string value toLowerCase(), toUpperCase(), length() and concat().

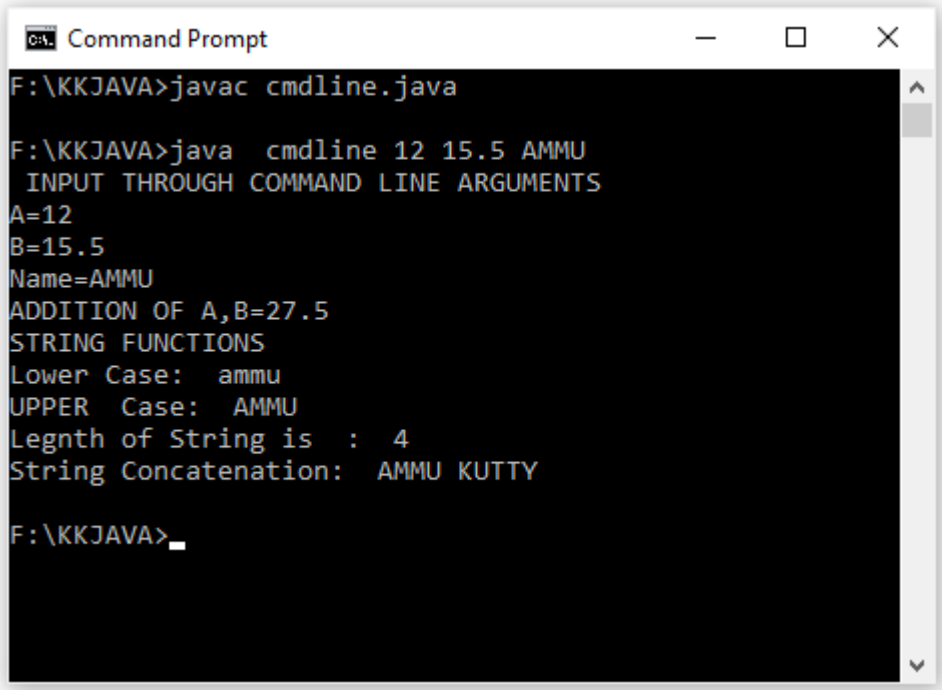
Step 7: stop the program.

**PROGRAM:**

**COMMAND-LINE ARGUMENTS AND STRING FUNCTIONS**

```
import java.io.*;
class cmdline
{
public static void main(String args[]) //args[0]=12 args[1]= 15.5 args[2]= AMMU
{
int a;
float
b,c;
String s;
a=Integer.parseInt(args[0]);
b=Float.parseFloat(args[1])
; s=args[2];
c=a+b;
System.out.println(" INPUT THROUGH COMMAND LINE ARGUMENTS");
System.out.println("A="+a); System.out.println("B="+b);
System.out.println("Name="+s);
System.out.println("ADDITION OF A,B=" + c);
System.out.println("STRING FUNCTIONS");
System.out.println("Lower Case: "+ s.toLowerCase());
System.out.println("UPPER Case: "+ s.toUpperCase());
System.out.println("Legnth of String is : "+ s.length());
System.out.println("String Concatenation: "+ s.concat(" KUTTY"));
}
}
```

**OUTPUT:**




**RESULT:**

The program takes command line arguments, converts them to respective data types, performs calculations, and implements string functions. The program runs successfully.

# INHERITANCE

**Ex. No. 10**
**Date:** 17.09.2025

**AIM:**

To implement the Inheritance concept using Java.

**ALGORITHM:**

Step 1: Start the program

Step 2: Create super class Room with methods

Step 3: Declare variable in the super class

Step 4: Create a sub class and extend the super class. Create a main class.

Step 5: Initialise variables in the main class

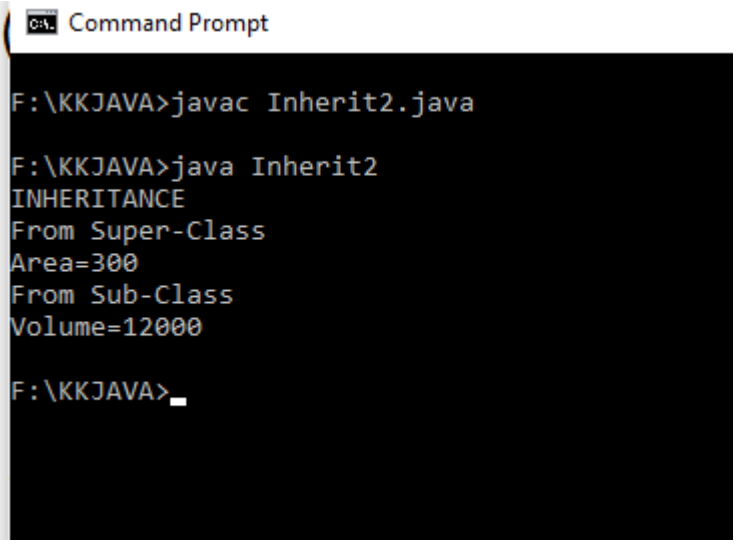
Step 6: Create an object for the subclass and call respective methods to perform the calculation.

Step 7: Display the output

**PROGRAM:**

```
import java.io.*;
class Room
{
intlength,width;
Room(int x, int y)
{
length=x;
width=y;
}
intarea()
{
return(length*width);
}
}
class PoojaRoom extends Room
{
int height; PoojaRoom(intx,int
y, int z)
{
super(x,y);
height=z;
}
intvolume()
{
return(length*width*height);
}
}

public class Inherit2
{
public static void main(String args[])
{
PoojaRoom P=new PoojaRoom(10,30,40);
int A=P.area();
int V=P.volume();
System.out.println("INHERITANCE");
System.out.println("From Super-Class");
System.out.println("Area="+A);
System.out.println("From Sub-Class");
System.out.println("Volume="+V);
} }
```

**OUTPUT:**




**RESULT:**

The program demonstrates inheritance in Java, where the PoojaRoom subclass extends the Room superclass. The program runs successfully.

# EXCEPTIONS

**Ex. No. 11**
**Date:** 24.09.2025

**AIM:**

To implement exceptions using Java.

**ALGORITHM**:

Step 1: Start the program

Step 2: Declare array A with size 3

Step 3: Take out array variables

Step 4: Display the output

Step 5: Stop the program

Step 6: Compile and run the program

Step 7: It show I/O exception array index out of bounds

Step 8: Stop the program

## 1. Division by Zero Exception

**PROGRAMS:**

```
import java.io.*; class
exceptiondemo
{
public static void main(String args[])
{
int a=10,b=5,c=5;
int x=a/(b-c);
System.out.println("X="+ x);
int y=a/(b+c);
System.out.println("Y="+ y);
}
}
```

## 2. Array Index Out of Bounds Exception

```
import java.io.*; class
exceptiondemo2
{
public static void main(String args[])
{
inta[ ]={10,20,30};
System.out.println("A[0]="+a[0]);
System.out.println("A[1]="+a[1]);
System.out.println("A[2]="+a[2]);
System.out.println("A[3]="+a[3]);
}
}
```

**OUTPUT-1**:

Command Prompt

```
F:\KKJAVA>javac exceptiondemo.java

F:\KKJAVA>java exceptiondemo
Exception in thread "main" java.lang.ArithmeticException: / by zero
        at exceptiondemo.main(exceptiondemo.java:9)

F:\KKJAVA>
```

**OUTPUT-2**:

Command Prompt

```
F:\KKJAVA>javac exceptiondemo2.java

F:\KKJAVA>java exceptiondemo2
A[0]=10
A[1]=20
A[2]=30
Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: 3
        at exceptiondemo2.main(exceptiondemo2.java:12)

F:\KKJAVA>
```

**RESULT:**

Therefore, two examples for JavExceptions have been demonstrated and executed successfully.

# MULTI THREADING

**Ex. No. 12**
**Date:** 01.10.2025

**AIM:**

To implement Multi-Threading using Java.

**ALGORITHM:**

Step 1: Start the program

Step 2: Create Three classes and Extend Thread

Step 3: Declare variables and assign values for each Thread Step 4:

Create main class and objects for each class

Step 5: Start and Run the Threads through their objects Step 5:

Stop the program

**PROGRAM:**

```
import java.io.*;
class A extends Thread
{
        public void run()
        {
        for(inti=1;i<=5;i++)
        {
        System.out.println("value from a i="+i);
        }
        System.out.println("exit from A");
        }
}
class B extends Thread
{
        public void run()
        {
        for(inti=1;i<=5;i++)
        {
        System.out.println("value from b i="+i);
        }
        System.out.println("exit from B");
        }
}
class C extends Thread
{
        public void run()
        {
        for(inti=1;i<=5;i++)
        {
        System.out.println("value from c i="+i);
        }
        System.out.println("exit from C");
        }
}
classthreaddemo
{
public static void main(String args[])
{
new A().start();
new B().start();
new C().start();
}
}
```

**OUTPUT:**

```
value from b i=1
value from a i=1
value from a i=2
value from b i=2
value from c i=1
value from a i=3
value from c i=2
value from c i=3
value from b i=3
value from c i=4
value from a i=4
value from c i=5
value from b i=4
exit from C
value from a i=5
value from b i=5
exit from A
exit from B
```

**RESULT:**

Thus, the program has been executed and the output, obtained successfully.

# PACKAGE

**Ex. No. 13**
**Date:** 01.10.2025

**AIM:**

To implement packages using Java.

**ALGORITHM:**

Step 1: Create a new folder p2

Step 2: Inside p2, create a sub folder P1 and class C2

Step 3: Create a package p1 in class C1 inside folder p1

Step 4: Class c2 as a main class which imports package p1

Step 5: Perform the operation

Step 6: Display the output

Step 7: Stop the program

**PROGRAM-1: PACKAGE P1\C1.JAVA**

```
package p1; public
class c1
{
public void display()
{
int l=10;
int h=20;
System.out.println("I am from first Package"); int
area=l*h;
System.out.println("Area of the Rectangle="+ area);
}
}
```

**PROGRAM-2: FOLDER P2\C2.JAVA**

```
import p1.*; public
class c2
{
public static void main(String args[])
{
c1 obj=new c1();
obj.display();
System.out.println("I am in the main class"); int
a=30, b=40;
int c=a+b; System.out.println("A="
+a); System.out.println("B=" +b);
System.out.println("Addition of A,B=" +c);
}
}
```

**OUTPUT:**

```
Command Prompt

F:\KKJAVA>cd p2

F:\KKJAVA\p2>cd p1

F:\KKJAVA\p2\p1>javac c1.java

F:\KKJAVA\p2\p1>cd..

F:\KKJAVA\p2>javac c2.java

F:\KKJAVA\p2>java c2
I am from first Package
Area of the Rectangle=200
I am in the main class
A=30
B=40
Addition of A,B=70

F:\KKJAVA\p2>_
```

**RESULT:**

The program successfully demonstrates package usage in Java by calculating a rectangle's area and performing addition. It prints the results of the 'area' and 'addition' programs, showcasing a successful package implementation.

# INTERFACE

**Ex. No. 14**
**Date:** 08.10.2025

**AIM:**

To implement an interface using Java.

**ALGORITHM:**

Step 1: Start the program

Step 2: Create an interface I1with a variable.

Step 3: Create a class C1 and implement Interface

Step 4: Access the variable in interface I1 and use it in the main class to perform calculation.

Step 5: Create a main class with an object to call the methods in C1.

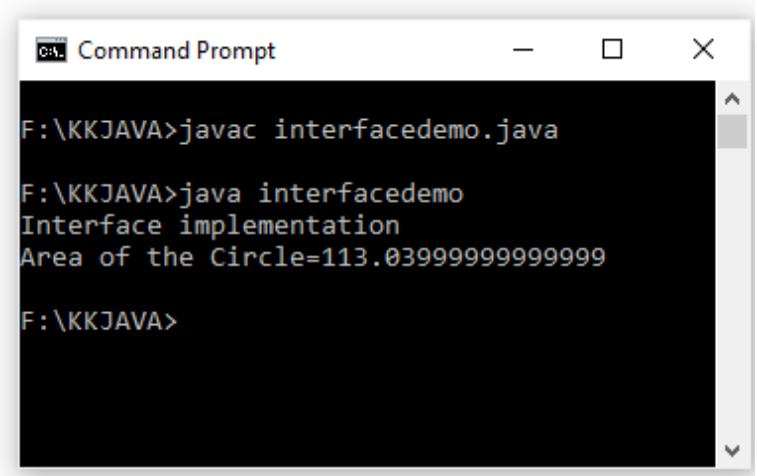
Step 6: Print the results.

Step 7: Stop the program

**PROGRAM:**

```
import java.io.*;
interface I1
{
int r=6;
}

class C1 implements I1
{
void display()
{
double pi=3.14;
System.out.println("Area of the Circle="+pi*r*r);
}
}
class interfacedemo
{
public static void main(String args[])
{
System.out.println("Interface implementation");
C1 obj=new C1();
obj.display();

}
}
```

**OUTPUT:**




**RESULT:**

The program implements an interface I1 with a variable r, which is used by class C1 to calculate and print the area of a circle. The output is Area of the Circle=113.04.

## LAYOUT MANAGER (FLOW LAYOUT)

**Ex. No.: 15**

**Date:** 08.10.2025

**AIM:**

To create a flow layout with a layout manager using Java.

**ALGORITHM:**

Step 1: Start the program

Step 2: Create a class and extend Frame

Step 3: Declare array S[]

Step 4: Create an object for Frame using Frame f= new Frame()

Step 5: Create Buttons and add buttons in flowlayout using FlowLayout()

Step 6: Create a main class and create an object for Frame class and show layout.

Step 7: Stop the program.

**PROGRAM:**

```
import java.awt.*;
import java.applet.*;
public class layout extends Applet
{
String s[]={"ONE","TWO","THREE","FOUR","FIVE","SIX"};
public void init()
{
for(inti=0;i<5;i++)
{
setLayout(new FlowLayout());
add(new Button(s[i]));
}
}
}
//<applet code="layout.class" height=400 width=500></applet>
```

**OUTPUT:**

Command Prompt

```
F:\KKJAVA>javac InputDemo.java
Note: InputDemo.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

F:\KKJAVA>java InputDemo
DATA INPUT STREAM
Enter a Numberic Value
50
Enter a Floating point Value
25.5
Enter a String Value
COMPUTER SCIENCE
A=50
B=25.5
S=COMPUTER SCIENCE
C=A+B:75.5

F:\KKJAVA>
```

**RESULT:**

The Java program creates a FlowLayout with buttons labeled "ONE" to "FIVE" using AWT. The buttons are arranged horizontally in a flow layout within an applet window.